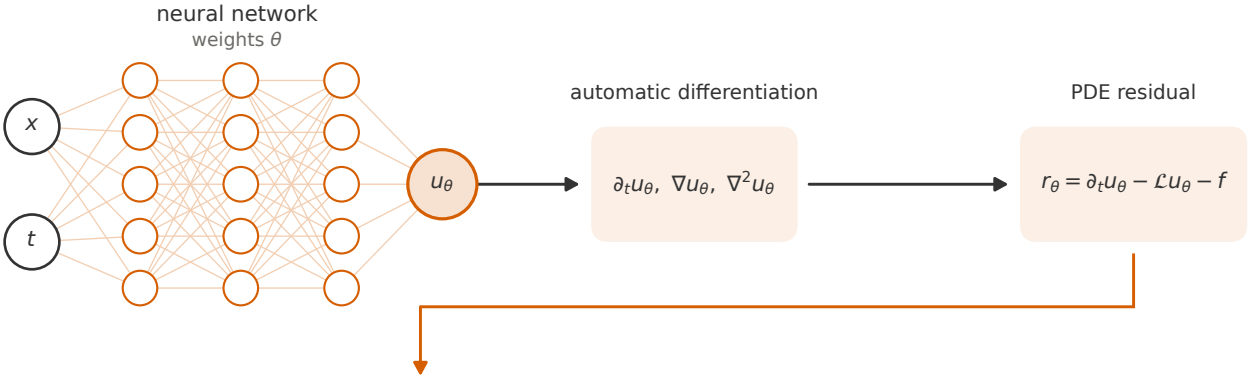
neural network
weights θ
x
t
u_θ
automatic differentiation
$\partial_t u_\theta,\ \nabla u_\theta,\ \nabla^2 u_\theta$
PDE residual
$r_\theta = \partial_t u_\theta - \mathcal{L} u_\theta - f$
$R(\theta) = \lambda_{\text{res}} R_{\text{res}} + \lambda_{\text{bc}} R_{\text{bc}} + \lambda_{\text{data}} R_{\text{data}}$
$\frac{1}{N}\sum_j r_\theta(x_j, t_j)^2$
$\frac{1}{N_{\text{bc}}}\sum_j (u_\theta - g)^2$
$\frac{1}{n_{\text{FK}}}\sum_i (u_\theta(x_i, t_i) - \bar{u}_i)^2$
PDE residual
boundary and initial data
Feynman–Kac targets